

AI-POWERED DELIVERY SYSTEM

AI-POWERED SELF-HEALING CI/CD

How a failed GitHub Actions run becomes an investigated, isolated, reviewable code fix.

A technical story in four components, one recovery loop.

THE SHIFT

FROM FAILURE ALERT TO PROPOSED FIX

The system keeps the judgment point with a reviewer while moving the repetitive investigation work closer to the failure.

TRADITIONAL PATH

- **Pipeline fails.** The signal waits for a developer.
- Developer investigates logs and source code.
- Developer fixes the code and opens a Pull Request.

THIS SYSTEM

Failure context arrives through FastAPI.

Google ADK investigates logs, jobs, and code.

A controlled change becomes a reviewable Pull Request.

ARCHITECTURE

FOUR COMPONENTS, ONE RECOVERY LOOP

Each component owns one boundary. Together they turn a failed run into a proposed change.

GITHUB ACTIONS

Runs the CI checks.

GITHUB APIS

Provide context and create the proposed change.



FASTAPI

Authenticates the failure webhook.

GOOGLE ADK

Orchestrates investigation.

THE CI PIPELINE CREATES THE FAILURE SIGNAL

The demonstrated workflow is deliberately narrow: it validates Python syntax before the larger recovery system acts.

Trigger

Pushes and Pull Requests targeting **main** or **master**.

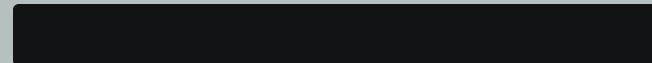
Runner

Ubuntu with Python 3.12 and dependency caching.

Install

Install the repository requirements.

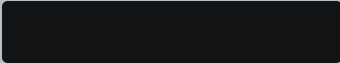
Quality gate



CURRENT SCOPE · Python syntax validation, not yet a full test suite.

FASTAPI TURNS A FAILURE INTO TRUSTED CONTEXT

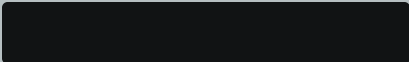
01 Receive

 carries repository, run ID, branch, head SHA, and optional logs.

02 Validate

Pydantic checks that the payload has the expected shape.

03 Authenticate

The Bearer token is checked against .

04 Enrich

Workflow jobs and logs are retrieved, with retry logic for temporary API delays.

THE ADK AGENT INVESTIGATES BEFORE IT EDITS

Google ADK coordinates an evidence-first investigation: map the repository, locate the failure, then read enough code to form a logical fix.



Map



Locate



Relate

Fetch utilities and configuration when dependencies matter.



Explain

Use logs plus code context to generate a

THE FIX IS ISOLATED, TRACEABLE, AND REVIEWABLE

01

Anchor

Create a branch from the exact failed commit identified by

02

Identify

Get the target file's Git blob SHA.

03

Commit

Base64-encode and commit the corrected source through

04

Review

Create a Pull Request into **main** with the problem, fix, and modified files.

The agent proposes the change. The reviewer decides whether it merges.



THE MEMORY DEVICE

THE TOLLYWOOD ANALOGY MAKES THE SYSTEM MEMORABLE

Repository is the movie project.

CI pipeline is the shooting process.

Failure logs are the shooting report.

FastAPI is the production manager and security gate.

ADK agent is the senior technician who fixes the failed scene.

Branch + Pull Request is the alternate movie version sent to the director.

THE PAYOFF

SELF-HEALING MEANS REMEDIATION, NOT JUST ALERTING

The system moves beyond “your pipeline failed” toward “I found the likely cause, prepared a fix, and opened a Pull Request for review.”

01

Autonomous investigation

Logs, jobs, repository structure, and source context are assembled before editing.

02

Controlled code change

The correction is isolated on a branch tied to the failed commit.

03

Human approval before merge

The Pull Request keeps the final decision with a reviewer.

INTERVIEW CLOSE · 30 SECONDS

ONE FAILURE. ONE PROPOSED PATH FORWARD.

GitHub Actions detects the failure. FastAPI authenticates and forwards the context. Google ADK coordinates the investigation. GitHub tools inspect and change code on an isolated branch. A Pull Request keeps the final decision with a reviewer.

Detect → authenticate → investigate → isolate → review

